BM4Midpoint: Arithmetic Diagonal Projection

GC2D numerical methods

Explicit palindromic operations

All ABBA and BM4 presets inherit `PalindromicMethod`. For signed $d = ch$, the elementary operations are

$$A_d(u, v, t) = (u + df(t, v), v, t), \qquad B_d(u, v, t) = (u, v + df(t, u), t), \qquad T_d(u, v, t) = (u, v, t + d).$$

Only T advances time. A coefficient $c = 1/2$ means duration $h/2$; there is no additional half inside A or B. The optional exact C flow rotates the copy difference and requires an independent coupling object. Both BM4 presets omit C.

In chronological order, standard ABBA2 is $A_{h/2}, B_{h/2}, T_{h/2}, T_{h/2}, B_{h/2}, A_{h/2}$. The distinct symmetric variant $T_{h/2}, A_{h/2}, B_{h/2}, B_{h/2}, A_{h/2}, T_{h/2}$ evaluates every spatial shear at the midpoint time. Both exhibit second order on the tested nonautonomous rotation. Arbitrary palindromes do not inherit fourth or sixth order, and an arithmetic projection need not be reversible or symplectic.

`Composition.from_half` reflects a supplied half sequence. The generic `PalindromicImplicit` and `PalindromicMidpoint` methods select one exterior Hairer projection or a final arithmetic mean. The standard signed ABBA/BM4 recipes, grouped observations and passive momentum normalization remain unchanged. NumPy and JAX execute the same explicit operations and differentiate their retained shear inputs. See `docs/models/extended/simulation/palindromic-operations.md`.

Current uncoupled configuration

Both `BM4Implicit` and `BM4Midpoint` omit the harmonic coupling flow and no longer accept a coupling-frequency constructor argument. Their twelve coefficients, signed clock and projection placement are unchanged. The independent operation `HarmonicCoupling` retains C for future compositions. Existing observer/archive rates are fixed at zero for new runs; nonzero historical calculations require their original frozen implementation.

1 Complete-cycle definition

For $\dot{z} = f(t, z)$, embed the accepted state as $Ez = (z, z)$ and define $P(u, v) = (u + v)/2$. One step is

$$z_{n+1} = P\Psi_h(Ez_n),$$

where Ψ_h is the same unprojected twelve-stage BM4 composition used by BM4Implicit. No projection is applied between stages and no multiplier is solved. "Midpoint" refers to the final arithmetic mean.

The coefficient vector is $(b_1, \ldots, b_6, b_6, \ldots, b_1)$, where

$$b_1 = 0.0792036964311957, \qquad b_2 = 0.1303114101821663,$$
$$b_3 = 0.2228614958676077, \qquad b_4 = -0.3667132690474257,$$
$$b_5 = 0.3246481886897062, \qquad b_6 = 0.1096884778767498.$$

The sum is one to floating-point accuracy. Starting with the adjoint, stages alternate adjoint and direct maps with signed duration $d_j = b_j h$. If $s_j = t_n + \sum_{i<j} d_i$, an adjoint stage uses s_j and a direct stage uses $s_j + d_j$. The negative coefficient also moves the stage clock backwards.

At the chosen evaluation time, let

$$A_d(u, v) = (u + df(t, v), v), \qquad B_d(u, v) = (u, v + df(t, u)).$$

The spatial direct map is $A_d \circ B_d$ and its adjoint is $B_d \circ A_d$. Neither includes harmonic coupling. The time translations remain in the complete direct/adjoint stage as specified above. Each stage evaluates the vector field twice. Thus one complete step uses 24 evaluations and one arithmetic projection.

2 State and energy conventions

Twelve direct/adjoint stages followed by one spatial arithmetic mean. Energy tracking is optional and passive. The two shear contributions in each stage update a summed momentum. Dividing this accumulator by two gives the physical kappa normalization. The common state conventions are specified below.

3 Order and geometric limitations

On the invariant diagonal of the exact extended flow, the two copies follow the original dynamics and the harmonic coupling vanishes. For a smooth field, the fourth-order unprojected composition therefore satisfies

$$\Psi_h(Ez) = E\varphi_h(z) + O(h^5).$$

Since $PE = I$ and P is linear, the projected method has local error $O(h^5)$ and global order four under the usual stability assumptions. The passive energy coordinates do not change the physical map or its order.

Arithmetic projection discards the copy difference. Consequently, symmetry or symplecticity of a base composition alone does not establish exact reversibility or symplecticity of the accepted physical map. This method makes neither claim. The recorded infinity norm $\|u_f - v_f\|_\infty$ measures the discarded separation before projection.

4 Executable checks

The test suite compares the physical map with an independent signed-stage replay and checks fourth-order convergence against an exact rotation and a tightly integrated nonautonomous reference with energy tracking both disabled and enabled. It also checks momentum normalization against an augmented reference ODE, particle batching, persistent observer callbacks, and independence from off-grid output sampling. Nonlinear iterations are absent by construction.

5 Four state formulations and passive energy tracking

For one planar particle the supported internal representations are

physical	$z = (x, y)$	$\mathbb{R}^2$,
physical with energy	(z, t, κ)	$\mathbb{R}^4$,
duplicated	$(u, v), \quad u, v \in \mathbb{R}^2$	$\mathbb{R}^4$,
duplicated with energy	(u, v, t, κ)	$\mathbb{R}^6$.

Classical methods use the physical rows; ABBA and BM4 use the duplicated rows. `track_energy` selects the energy variant. For N planar particles the respective dimensions are $2N$, $4N$, $4N$, and $6N$: one time entry and one momentum per particle. All time entries equal the integration time. Classical full-cyclotron runs instead use $4N$ or $6N$ entries. Accepted spatial copies coincide; only one physical copy is exported. Every coordinate block contains N values in component-major order. The formulation validates particle times and aligns output round-off; the integrator does not interpret coordinate indices.

The normalized passive balance is

$$\dot{\kappa} = -\partial_t H(t, z), \qquad \kappa(t_0) = 0, \qquad \Delta K(t) = H(t, z(t)) + \kappa(t) - H(t_0, z_0).$$

Splitting accumulators from both copies are divided by two. The physical Hamiltonian may vary with time; ΔK measures its computed balance. Tracking leaves physical trajectories, nonlinear stopping rules and adaptive acceptance unchanged. Dynamics must supply the momentum derivative, including zero for an autonomous Hamiltonian.

Hairer's constraint is solely $u - v = 0$: the reduced multiplier has $2N$ entries and ABBA's simultaneous spatial solve has $6N$ unknowns. Neither t nor κ is projected. The former eight-component projection is retired; `state_extension='fully_extended'` raises migration guidance.

Dynamics, formulation, method and integration remain separate. The shared state classes own layouts, diagonal embedding and history extraction in

```
src/formulations/state.py
```

Each integration initializes a fresh method instance. Observers receive physical states; energy histories are in `Solution.diagnostics`. The `extended_time` history has shape (N, n_{samples}). Fixed-step shadow samples leave main states and counters unchanged. Adaptive energy quadrature uses accepted dense output; its backend state and Jacobian remain physical.

6 Shared extended-space implementation

Public method names remain unchanged; both BM4 constructors now omit harmonic coupling. Their code lives in `src/methods/extended/`. Public exports from `simulation` remain available; former ABBA and BM4 import adapters have been removed. The shared execution is `advance`, `solve_projection`, `compose`, then explicit $A/B/T/C$ operations. Arithmetic midpoint variants replace the nonlinear projection by one final average.

The composition retains signed shear inputs for exact particle-block tangents, passive energy quadrature, and existing observation records. Newton and Broyden share the spatial projection equations; energy is accumulated only from the accepted trace, with the existing factor 1/2 for physical κ. Adjoint stages evaluate at the start of their signed substep, direct stages at its end. The coefficients remain unchanged. ABBA4, ABBA6 and BM4 each use one outer projection around their complete unprojected compositions. ABBA6 therefore differs from historical results with

seven projected pairs. This internal family name does not enable the retired full time/momentum projection.

The implementation architecture and current diagrams are documented in `docs/models/extended/simulation/extended-simulation-architecture.md`.

7 Optional JAX execution

An optional JAX driver executes the same numerical map on one selected CPU or GPU, with particle operations batched inside a compiled sequential time loop. It reuses the canonical coefficients and prepared potential splines. Physical-state tolerance scaling, signed stage times, passive energy weights and independent shortened output maps retain their definitions above. Applicable Newton systems use independent particle blocks; an explicitly selected Broyden iteration retains its global secant matrix. Nonlinear iteration limits and failure checks remain active on the device.

Both execution routes use double precision and return the same physical solution layout. Floating-point reassociation can produce round-off differences without changing the designed order or geometric properties. First-call compilation is separate from warmed integration timings, which include transfers and output construction. The sibling runtime document and shared execution contract specify supported callbacks and exact device scope.